MIL-I-82681(OS)
30 October 1978
Superseding
(see 6.4)

MILITARY SPECIFICATION

INGOT, BISMUTH-TIN ALLOY, EUTECTIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the eutectic alloy of bismuth and tin in the form of ingots.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 9650

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8-77 Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, Washington, DC 20026).

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material and dimensions. The material shall be the eutectic alloy of bismuth and tin formed into solid ingots. Ingot dimensions shall be a minimum of 0.75 X 1.25 X 5.50 inches.

3.2 Physical and chemical properties. Physical and chemical properties of the material shall conform to the requirements shown in TABLE I.

TABLE I. Physical and chemical properties.

Property	Value	
	Minimum	Maximum
Tensile strength, psi	9500	-
Bismuth content, wt%	56	60
Tin content, wt%	40	44
Freezing temperature, °C	136	139

3.3 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to perscribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: No requirement
d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Quality conformance inspection. The quality conformance inspection shall consist of the tests specified in 4.4.

4.3.1 Lot. A lot shall consist of material from a single batch compounded without changes in process or materials and to be offered for acceptance at one time.

4.3.2 Sampling. Ten percent, but in no case more than 10 nor less than 2 of the ingots shall be randomly selected by the inspector to be representative of the lot. If there is one ingot in the lot, it shall be sampled. Specimens for tensile strength shall be prepared from each sample ingot. After testing for tensile strength, the pieces shall be composited and tested for bismuth content, tin content, and freezing temperature.

4.4 Test methods. The following tests shall be performed on each lot. Failure of any sample to conform to any of the requirements of this specification shall be cause for rejection of the lot. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218. All weights shall be recorded to the nearest 0.1 milligram (mg).

4.4.1 Tensile strength. Tensile strength shall be determined in accordance with ASTM E 8-77 under the following conditions:

a. The test specimen dimensions shall be in accordance with the 0.350 inch round specimens shown in FIGURE 8 of ASTM E 8-77.

b. The rate of stress shall be 4000 pounds per square inch per minute.

c. The testing temperature shall be 24 ± 3°C (75 ± 5°F).

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4.4.2 Bismuth content. Bismuth content shall be determined in accordance with the following procedure:

a. Procedure. Weigh approximately 0.5 g of sample shavings, previously cleaned with toluene, in a 250-milliliter (mL) beaker. Add 10 mL of 1:1 nitric acid and place on a steam bath for 1 hour. Dilute with approximately 75 mL of water and digest for 1 hour on a steam bath. Filter while hot, wash with approximately 25 mL of 1:1 nitric acid, and collect filtrate in a 400-mL beaker. Puncture the filter paper and wash the precipitate into the original 250-mL beaker with approximately 25 mL of 1:1 nitric acid. Dilute with approximately 75 mL of water, digest for 1 hour on the steam bath, and refilter into the 400-mL beaker containing the original filtrate. Neutralize the combined filtrates with concentrated ammonium hydroxide until a faint turbidity is observed. Add 2 mL of 1:3 hydrochloric acid and digest for 2 hours or until precipitate settles. Cool and filter through a tared, medium porosity, sintered glass crucible. Dry in an oven and weigh. Report the average of a minimum of two determinations.

b. Determine the constant of recovery (K) in accordance with procedure a. on a standard consisting of 0.29 g of bismuth and 0.21 g of tin.

c. Calculation. $K = \frac{(P)\ (0.8023)}{B}$

where:
P = weight of precipitate from standard, g
0.8023 = conversion factor, bismuth oxychloride to bismuth
B = weight of bismuth used in standard, g

d. Calculation. $\text{Bismuth (wt\%)} = \frac{(P)\ (0.8023)}{(W)\ (K)}$

where:
P = weight of precipitate from unknown, g
0.8023 = conversion factor, bismuth oxchloride to bismuth
W = weight of sample, g
K = constant of recovery (from c.)

4.4.3 Tin content. Tin content shall be determined in accordance with the following procedure:

a. Procedure. Weigh approximately 0.5 g of sample shavings, previously cleaned with toluene, in a 250-mL beaker. Add 10 mL of concentrated hydrochloric acid, cover beaker with watch glass, place on hot plate, and allow to boil freely. When the volume is reduced to approximately 10 mL, add an additional 10 mL of hydrochloric acid. Repeat this treatment twice. Transfer the contents of the beaker to a 200-mL flask equipped with a side

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arm for introduction of carbon dioxide. Add 25 mL of concentrated hydrochloric acid and 5 g of nickel. Connect the carbon dioxide line to the flask, connect to condenser, and place flask on a hot plate. Boil for approximately 40 minutes, remove from hot plate, and cool. With the carbon dioxide line connected, titrate with 0.1 normal (N) iodine, using starch indicator, to first appearance of blue color. Report the average of a minimum of two determinations.

b. Determine the apparent milliequivalent weight (K) in accordance with procedure a. on a standard consisting of 0.29 g of bismuth and 0.21 g of tin.

c. Calculation. $K = \frac{T}{(V)\ (N)\ (1000)}$

where: T = weight of tin used in standard, g
V = volume of iodine used in titrating standard, mL
N = normality of iodine solution
1000 = conversion factor, equivalents to milliequivalents

d. Calculation. Tin content (wt%) $= \frac{(V)\ (N)\ (K)}{W} \times 100$

where: V = volume of iodine used in titrating unknown, mL
N = normality of iodine solution
K = apparent milliequivalent weight (from c.)
W = weight of sample, g

4.4.4 Freezing temperature. Freezing temperature shall be determined in accordance with the following procedure:

a. Procedure. Place approximately 200 g of sample in a pyrex test tube. Heat slowly until the sample melts. Cautiously introduce a thermometer into the melted sample and continue heating until the sample temperature is 150 ± 5°C. Skim the surface of the sample. Place the test tube, with sample and thermometer, in an insulating jacket and assemble the apparatus as shown in Figure 1. As the temperature begins to fall, agitate with the stirrer. Record the temperature every 30 seconds. Stir moderately as long as the sample is fully molten. Allow the thermometer to freeze in this sample. The sample will supercool, and then as crystallization begins, the temperature will rise to a plateau which is maintained for approximately 2 minutes. The temperature corresponding to the plateau, corrected for stem effect, shall be recorded as the freezing temperature.

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4.5 Packaging, packing, and marking. The packaging, packing, and marking shall be examined for conformance to this specification.

PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be commercial.

5.1.1 Commercial. Unless otherwise specified in the contract (see 6.2), packaging and packing of the bismuth-tin alloy ingot shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and location.

c. Material trade name.

d. Net weight and volume.

e. Lot number, batch number(s), and date of manufacture.

f. Storage conditions.

g. Contract number.

6. NOTES

6.1 Intended use. The material is intended for use as a fusible plug in the Mk 12 missile booster for the Standard Missile.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification

b. Place of delivery

c. Quantity required

d. Inspection condition, if other than as specified (see 4.2)

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e. Packaging and packing information, if other than as specified (see 5.1)

f. Any special marking required

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD 1664) which will be documented in the Applicable ADL (Authorized Data List).

6.3 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is Cerrotru Alloy manufactured by Cerro Metal Products Division of Cerro-Marmon Corporation, P. O. Box 444, 6707 Mowry Ave., Newark, CA 94560. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

6.4 Supersession information. MIL-I-82681 includes the requirements of and supersedes Naval Sea Systems Command (Code Ident 10001) purchase description WS2517 dated 25 March 1966.

Custodian:
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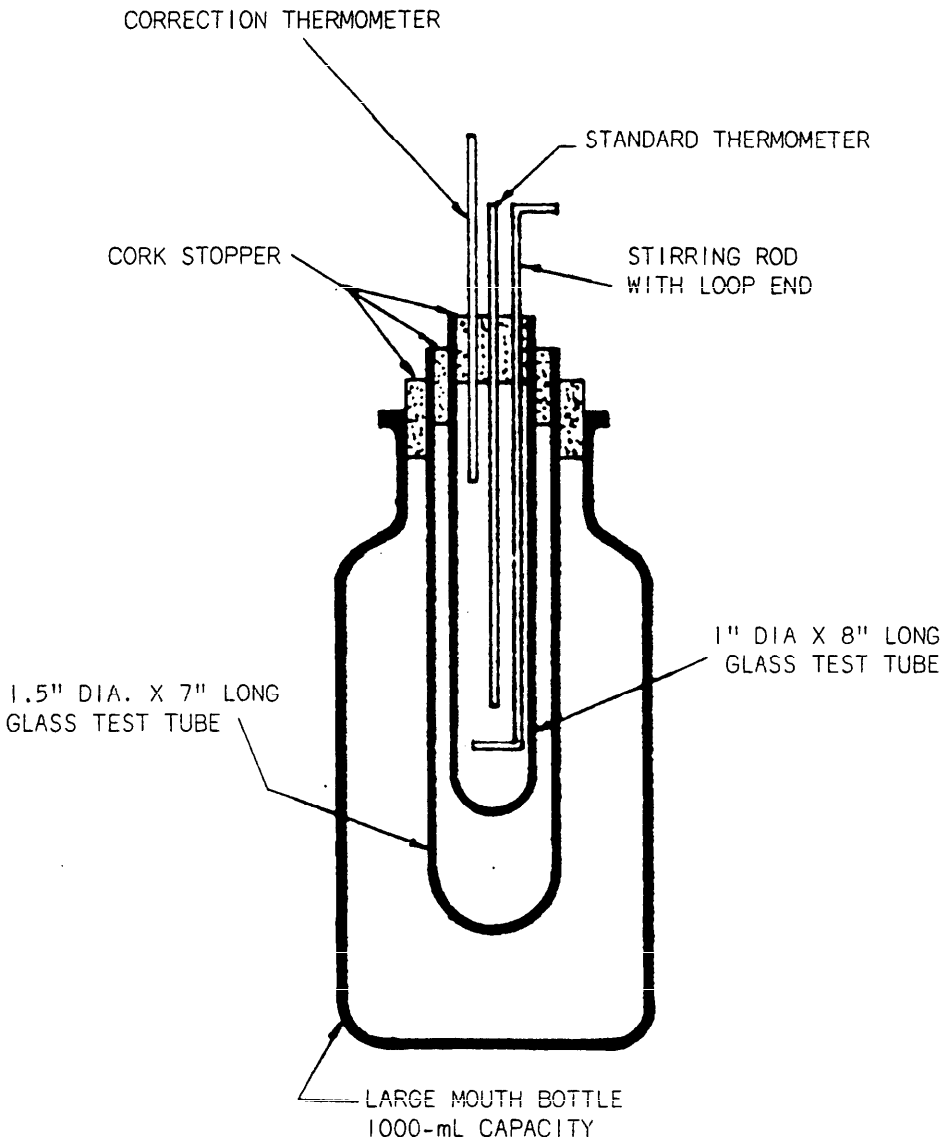


FIGURE 1. Freezing temperature apparatus.

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DOCUMENT IDENTIFIER AND TITLE
MIL-I-82681(OS), INGOT, BISMUTH-TIN ALLOY, EUTECTIC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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